MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, DIFFERENTIAL, ANALOG, KEY CODE FGH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic differential amplifier module.

1.2 Classification. This specification covers two classes of modules.

Class I and Class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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* 3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.183 failures per million operating hours for class I or 0.622 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FGH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/269-1 for class I and M28787/269-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of ±15.0 ±1.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of ±16.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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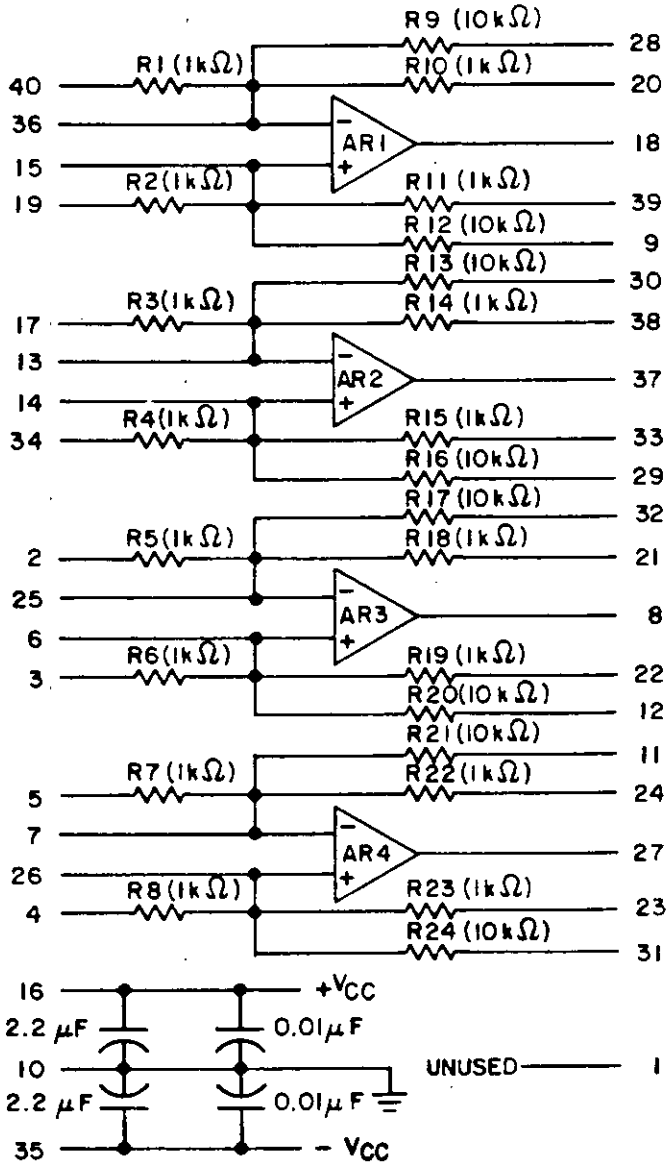


FIGURE 1. FGH module function diagram, differential amplifier.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A4	V_S stress	II	Q	-90	90	--	--	--	--	--	--	--	--	mV
B1-B8	Output short circuit current	III	S	9	40	9	55	9	40	9	55	9	40	mA
C1,C2	Isolation	IV	S											
	Initial			10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life			1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
D1,D2	Power supply current	V	S	1.0	12.4	1.0	13.0	1.0	12.0	1.0	14.0	1.0	11.0	mA
E1,E2	Power supply filter capacitance	VI	X	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	μF
F1-F16	Resistance	VII	X											
	Initial			0.990	1.010	0.985	1.015	0.985	1.015	0.981	1.019	0.984	1.016	kΩ
	End-of-life			0.985	1.015	0.980	1.020	0.980	1.020	0.976	1.024	0.976	1.024	kΩ
F17-F24	Resistance	VII	X											
	Initial			9.90	10.10	9.85	10.15	9.85	10.15	9.81	10.19	9.84	10.16	kΩ
	End-of-life			9.85	10.15	9.80	10.20	9.80	10.20	9.76	10.24	9.76	10.24	kΩ
G1-G4	Input offset voltage check	VIII	X	-3.0	3.0	-3.5	3.5	-3.5	3.5	-4.0	4.0	-4.0	4.0	mV
H1-H4	Input offset current check	VIII	X	-30	30	-43	43	-37	37	-70	70	-45	45	nA
J1-J4	Input bias current check	IX	X	1	110	1	265	1	110	1	265	1	110	nA
K1-K4	AC gain functional check (Gain = 10)	X	X											
	Initial			9.70	10.20	9.60	10.30	9.60	10.30	9.52	10.38	9.52	10.32	V/V
	End-of-life			9.60	10.30	9.50	10.40	9.50	10.40	9.43	10.48	9.43	10.42	V/V
K5-K8	AC gain functional check (Gain = 1)	X	X											
	Initial			0.98	1.02	0.97	1.03	0.97	1.03	0.962	1.038	0.962	1.038	V/V
	End-of-life			0.97	1.03	0.96	1.04	0.96	1.04	0.952	1.048	0.952	1.048	V/V
L1-L4	Small signal bandwidth	X	S	0.438	5.000	0.460	5.000	0.403	5.000	0.493	5.000	0.372	5.000	MHz

See rootnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
M1-M4	Gain peaking check	X	S	1.00	1.35	1.00	1.35	1.00	1.35	1.00	1.35	1.00	1.35	Ratio
N1-N4	Large signal voltage gain check	XI	X	50	700	25	700	25	700	25	700	25	700	V/mV
P1-P8	Output slew rate check	XII	X	0.30	10.0	0.29	10.0	0.30	10.0	0.24	10.0	0.30	10.0	V/µs
R1-R4	Maximum output voltage	XIII	S	6.00	9.55	6.00	9.55	6.00	9.55	6.00	9.55	6.00	9.55	V rms
S1-S8	Common mode rejection ratio check	XIV	X	-∞	-40	-∞	-40	-∞	-40	-∞	-40	-∞	-40	dB
T1-T4	Power supply rejection ratio check	XV	S	-∞	-52	-∞	-52	-∞	-52	-∞	-52	-∞	-52	dB
U1,U2	Circuit-to-circuit phase shift	XVI	S	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	-0.5	0.5	Degree
V1-V4	Crosstalk check	XVII	Q	-∞	-40	-∞	-40	-∞	-40	-∞	-40	-∞	-40	dB
W1-W4	Output latch	Fig. 4	Q	---	---	---	---	-61	61	---	---	-66	66	mV

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or W are part of a sequence of tests, the order of testing shall be as follows: W, A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B and D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4. Following the life test and prior to disconnecting the power, electrical test W shall be performed in accordance with figure 4. While still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Sustitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation ±16.5 V dc supply (no signal)	215 max	231 max	mW
Supply voltage	±15 ±10%	±15 ±10%	V dc

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	Class I	Class II	Unit
Input offset voltage	±3.5 max	±4.0 max	mV
Input offset current	±43 max	±70 max	nA
Input bias current (average, each input)	265 max	265 max	nA
Common mode rejection ratio	-40 max	-40 max	dB
Power supply rejection ratio	-52 max	-52 max	dB
Slew rate (unity gain) ($R_L \geq 2$ kΩ, V_{CC} = -15 V)	0.29 min	0.24 min	V/μs
Small signal bandwidth (unity gain)	0.403 min	0.372 min	MHz
Large signal voltage gain	25,000 min	25,000 min	---
Gain options (differential)	1,2,10,20	1,2,10,20	---
Gain accuracy (F = 1.0 kHz, $R_L \geq 2$ kΩ)	±5.2 max	±5.2 max	%
Gain peaking	≤1.35	≤1.35	Ratio
Output voltage (R_L = 2 kΩ, F = 1.0 kHz, V_{CC} = 13.5 V, <1% distortion)	6.00 max	6.00 max	V rms
Output drive current (V_{OUT} = 8.5 V_{peak})	4.25 min	4.25 min	mA

6.5 Module function. This module contains four independent differential amplifiers and 24 resistors for gain programming. Both amplifier inputs are brought out to module pins, thus allowing programming of other op-amp configurations.

6.5.1 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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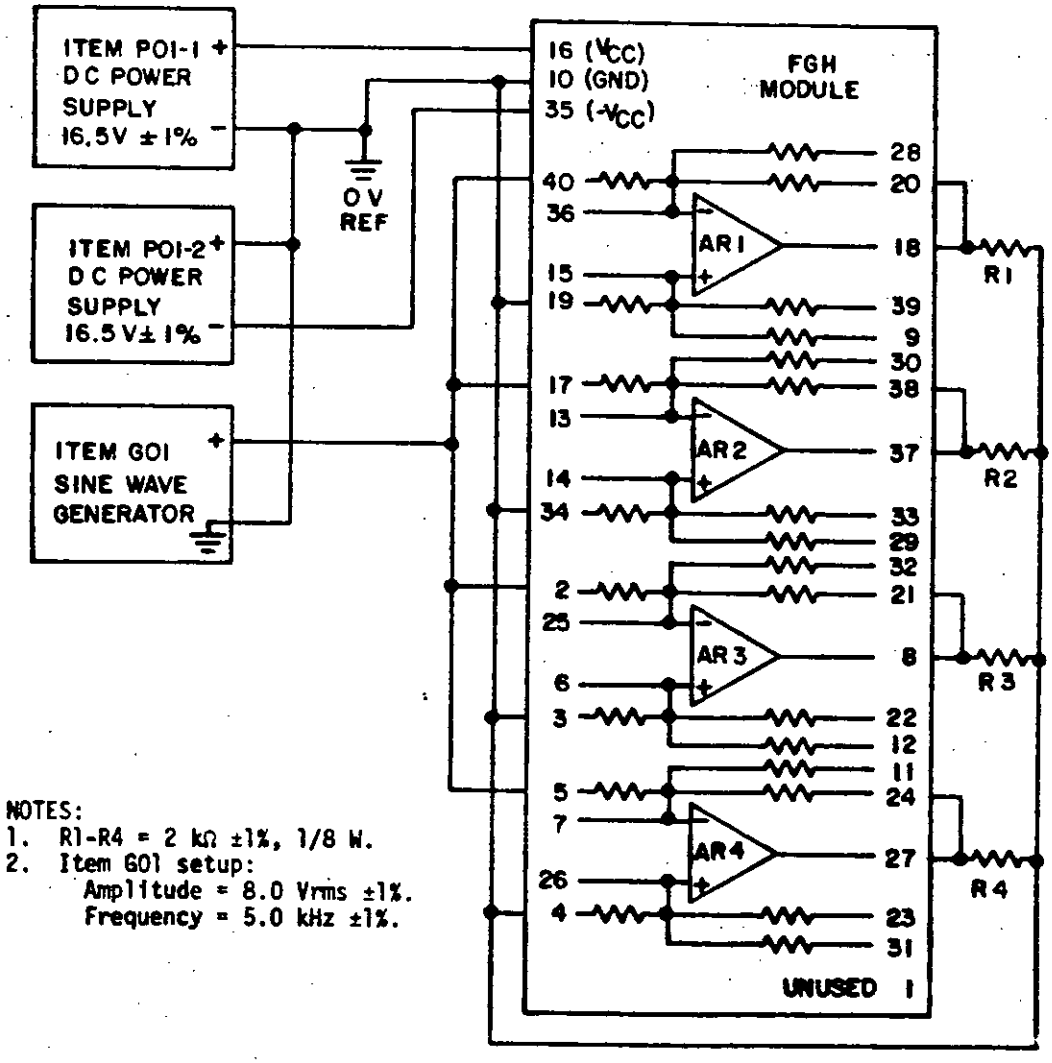


FIGURE 2. Operating temperature and component temperature test circuit.

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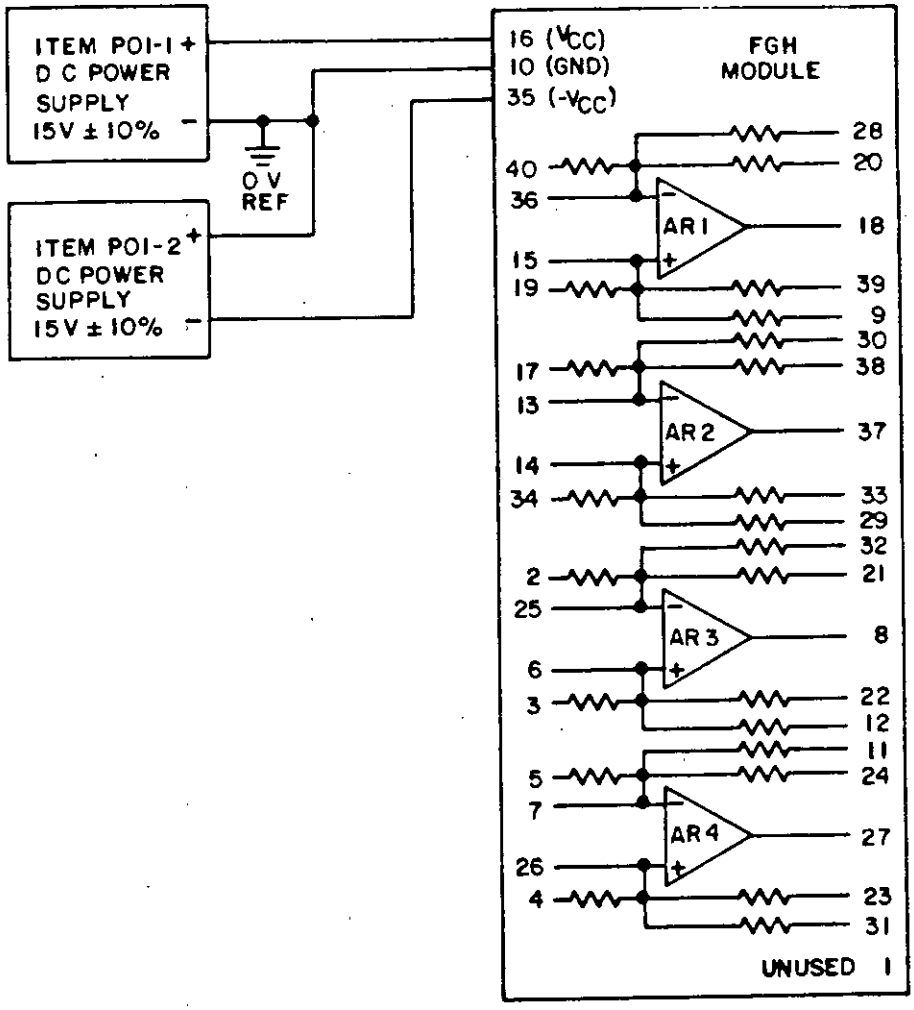
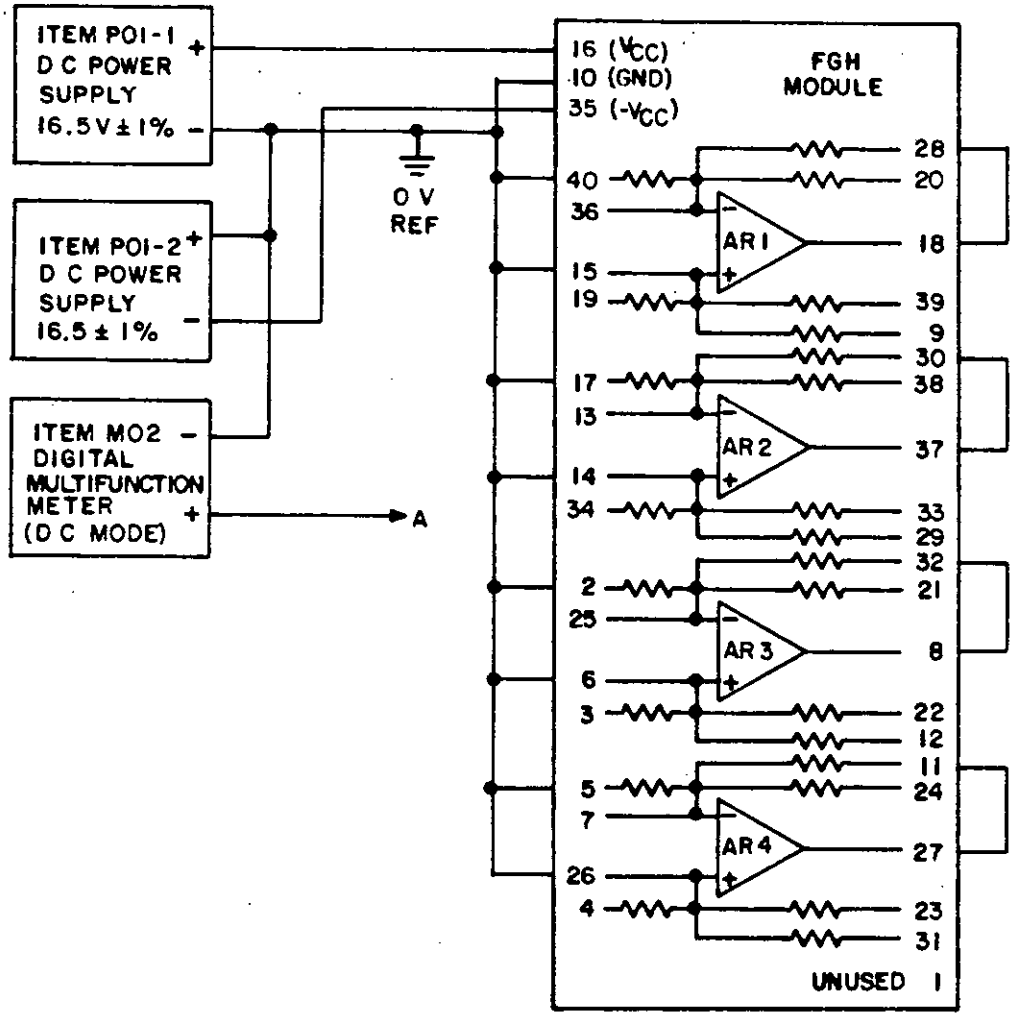


FIGURE 3. Vibration test circuit.

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NOTES:
1. Following the life test period and prior to disconnecting the power, connect point A to pin 18. The voltage shall meet the limits specified in table I for test condition W1.
2. For test conditions W2, W3, and W4, connect point A to pins 37, 8, and 27 respectively.
3. Test condition W verifies that the amplifiers do not latch up.

FIGURE 4. Life test circuit.

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TABLE II. V_S stress test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	V_S stress	18	-90	90	mV	5
A2	↓	37	↓	↓	↓	↓
A3		8				
A4		27				

TABLE III. Output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
B1	Output short circuit current	Open	9	18	10	9	40	mA	6
B2	↓	↓	29	37	↓	↓	↓	↓	↓
B3			12	8					
B4			31	27					
B5		9	Open	10	18				
B6		29	↓	↓	37				
B7		12			8				
B8		31			27				

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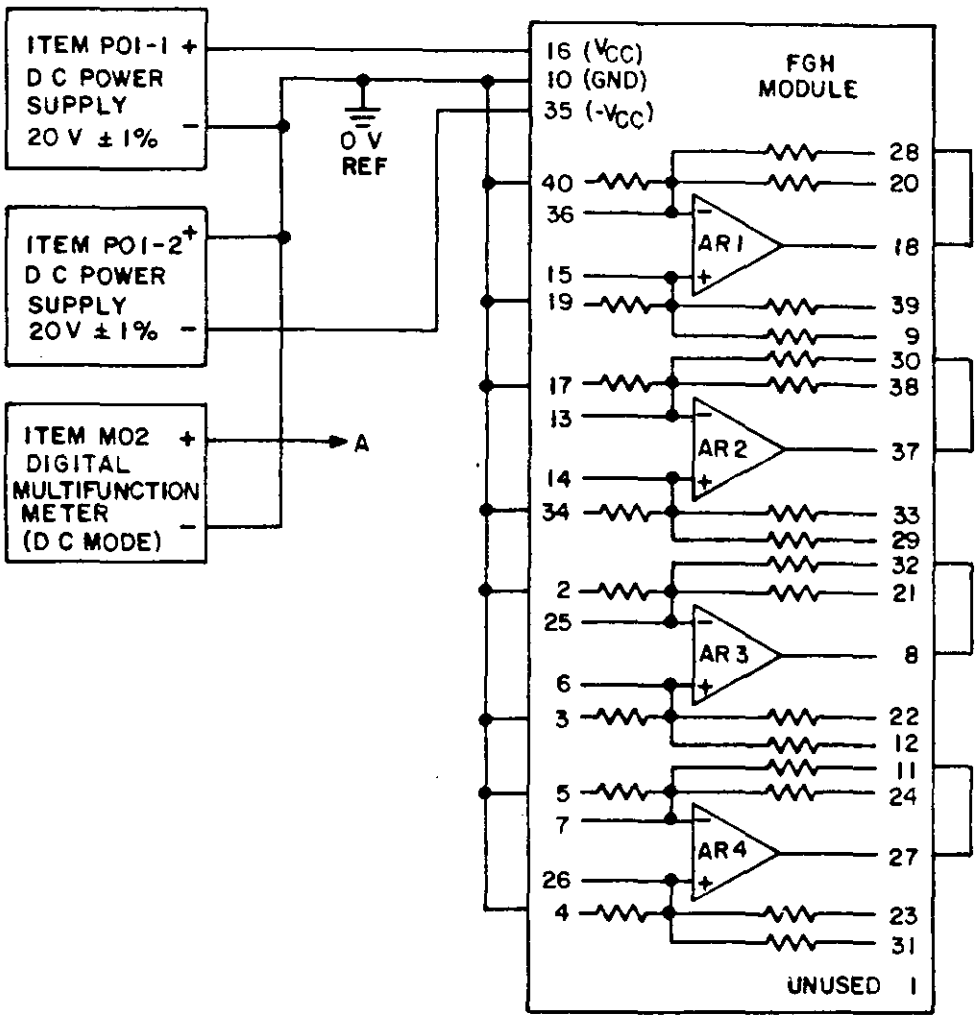
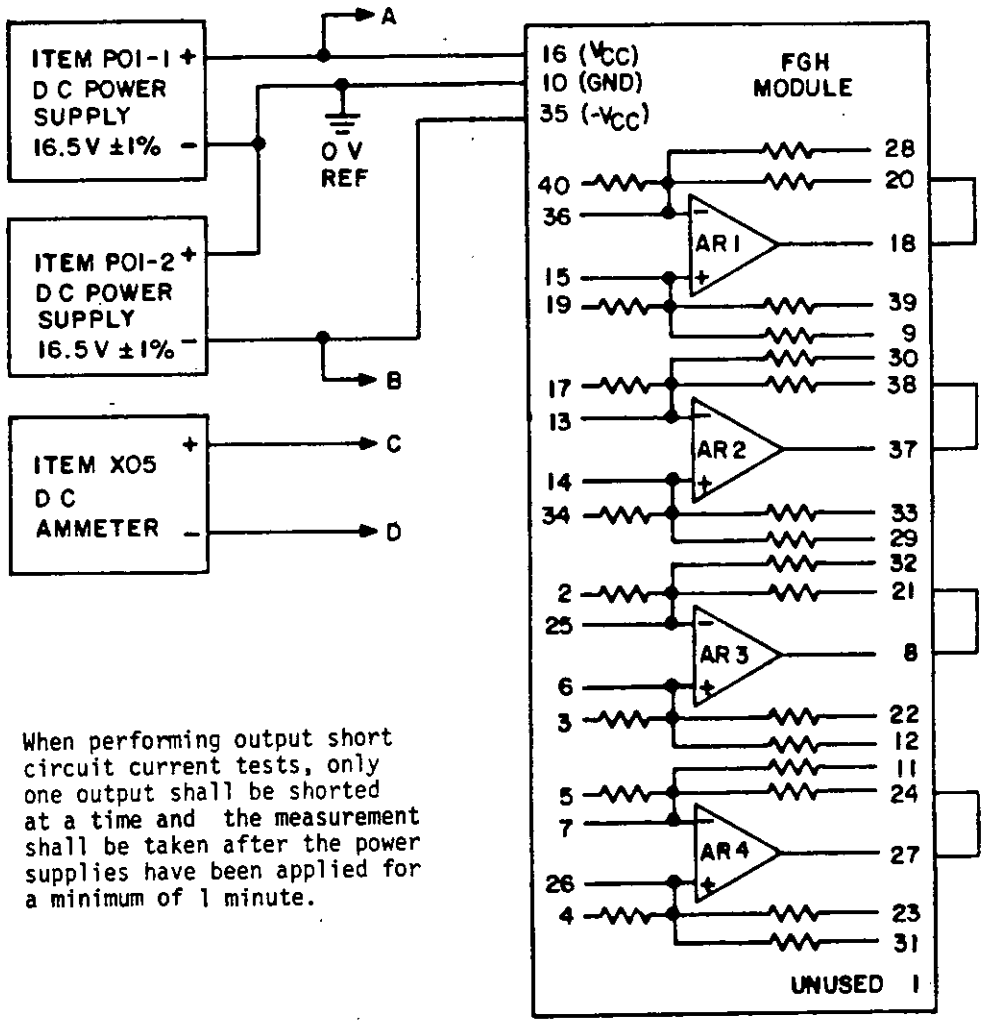


FIGURE 5. V_S stress test circuit for table II.

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NOTE: When performing output short circuit current tests, only one output shall be shorted at a time and the measurement shall be taken after the power supplies have been applied for a minimum of 1 minute.

* FIGURE 6. Output short circuit current test circuit for table III.

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TABLE IV. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation	1-40	Frame	10	∞	MΩ	7
C2	Isolation	2-40	1	10	∞	MΩ	7

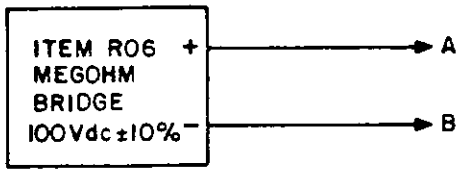
TABLE V. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
D1	Power supply current	1.0	12.4	mA	8
D2	Power supply current	1.0	12.4	mA	8

TABLE VI. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	Power supply filter capacitance	16	10	1.65	2.75	µF	9
E2	Power supply filter capacitance	10	35	1.65	2.75	µF	9

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WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition C1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of item R06.

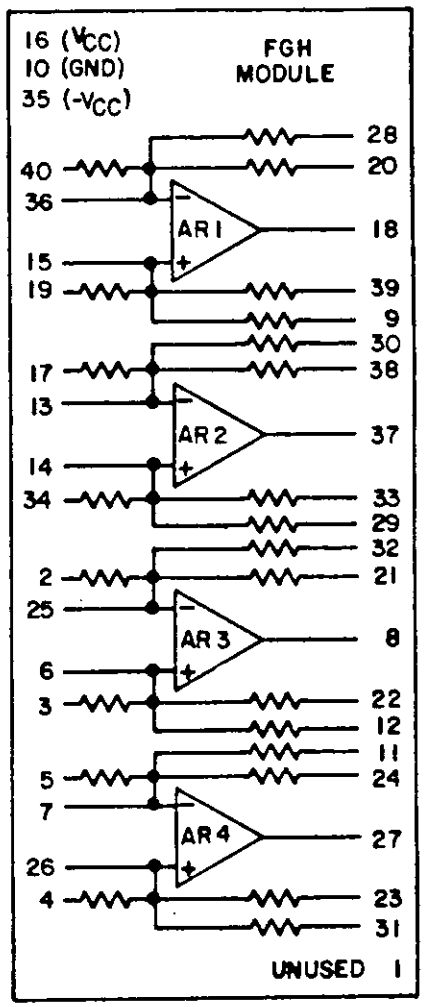
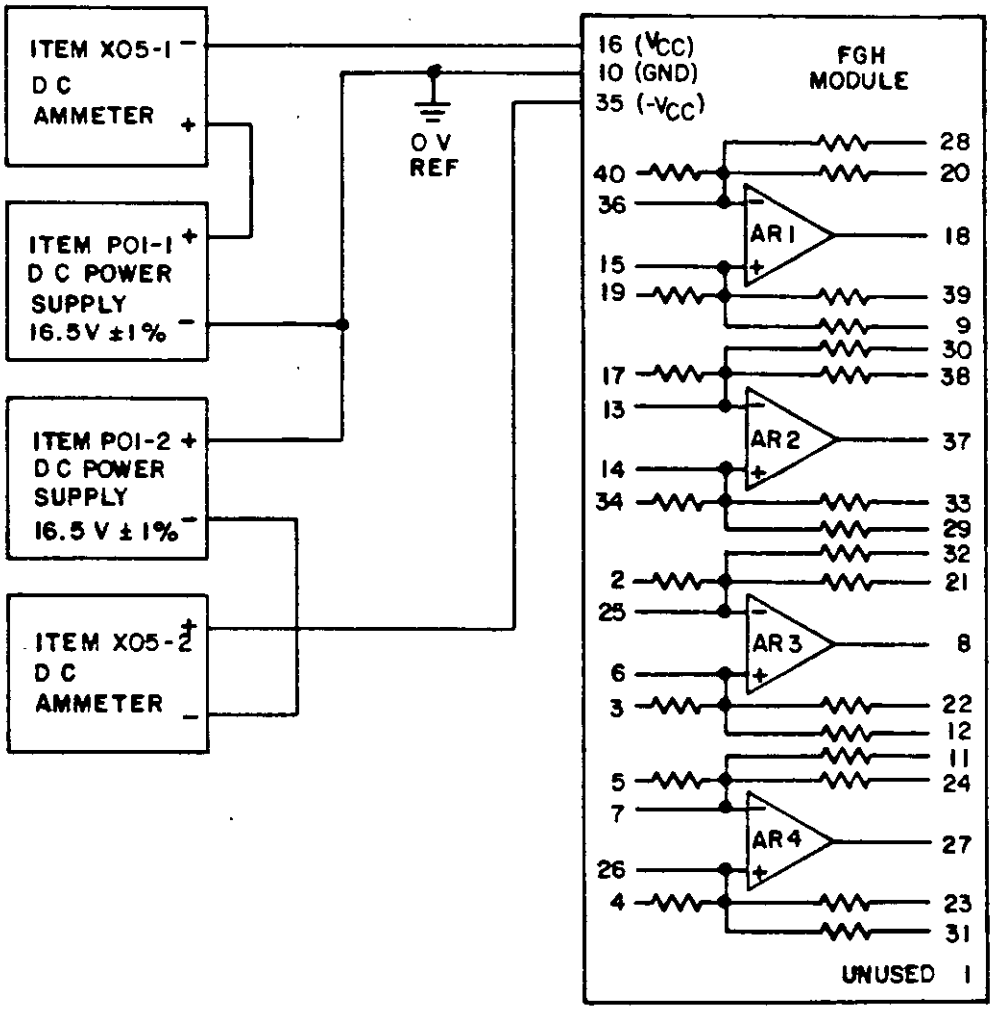


FIGURE 7. Isolation test circuit for table IV.

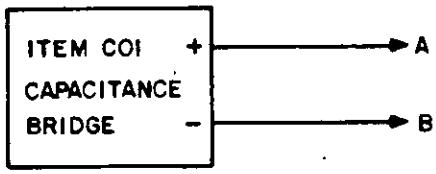
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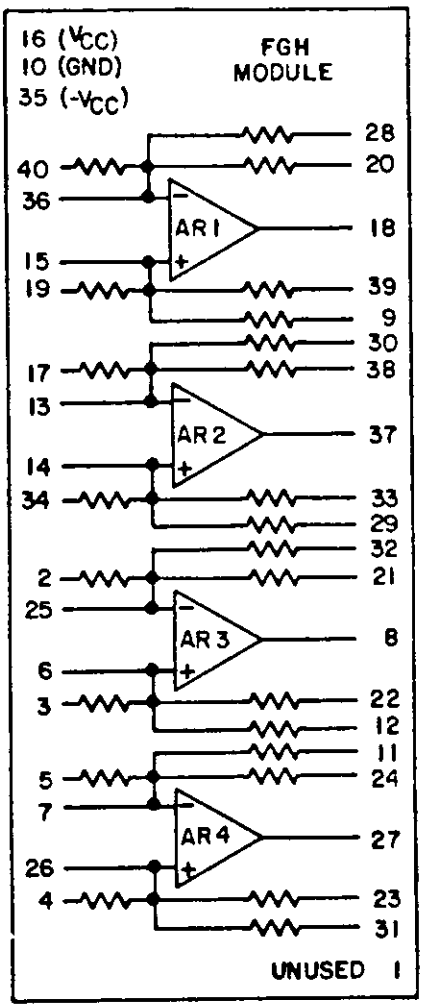


NOTES:
1. Read item XO5-1 for test condition D1 and item XO5-2 for test condition D2.
2. Test conditions D1 and D2 are not run simultaneously; therefore, only one item XO5 is needed.

FIGURE 8. Power supply current test circuit for Table V.

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NOTE: Capacitance bridge C01 setup:
Frequency = 1 kHz ±10%.
Amplitude = 0.3 Vpk ±10%.

* FIGURE 9. Power supply filter capacitance test circuit for table VI.

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TABLE VII. Resistance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Resistance	40	36	0.990	1.010	kΩ	10
F2		19	15				
F3		17	13				
F4		34	14				
F5		2	25				
F6		3	6				
F7		5	7				
F8		4	26				
F9		20	36				
F10		39	15				
F11		38	13				
F12		33	14				
F13		21	25				
F14		22	6				
F15		24	7				
F16		23	26				
F17		28	36	9.90	10.10		
F18		9	15	9.90	10.10		

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TABLE VII. Resistance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F19	Resistance	30	13	9.90	10.10	kΩ	10
F20	↓	29	14	↓	↓	↓	↓
F21		32	25				
F22		12	6				
F23		11	7				
F24		31	26				

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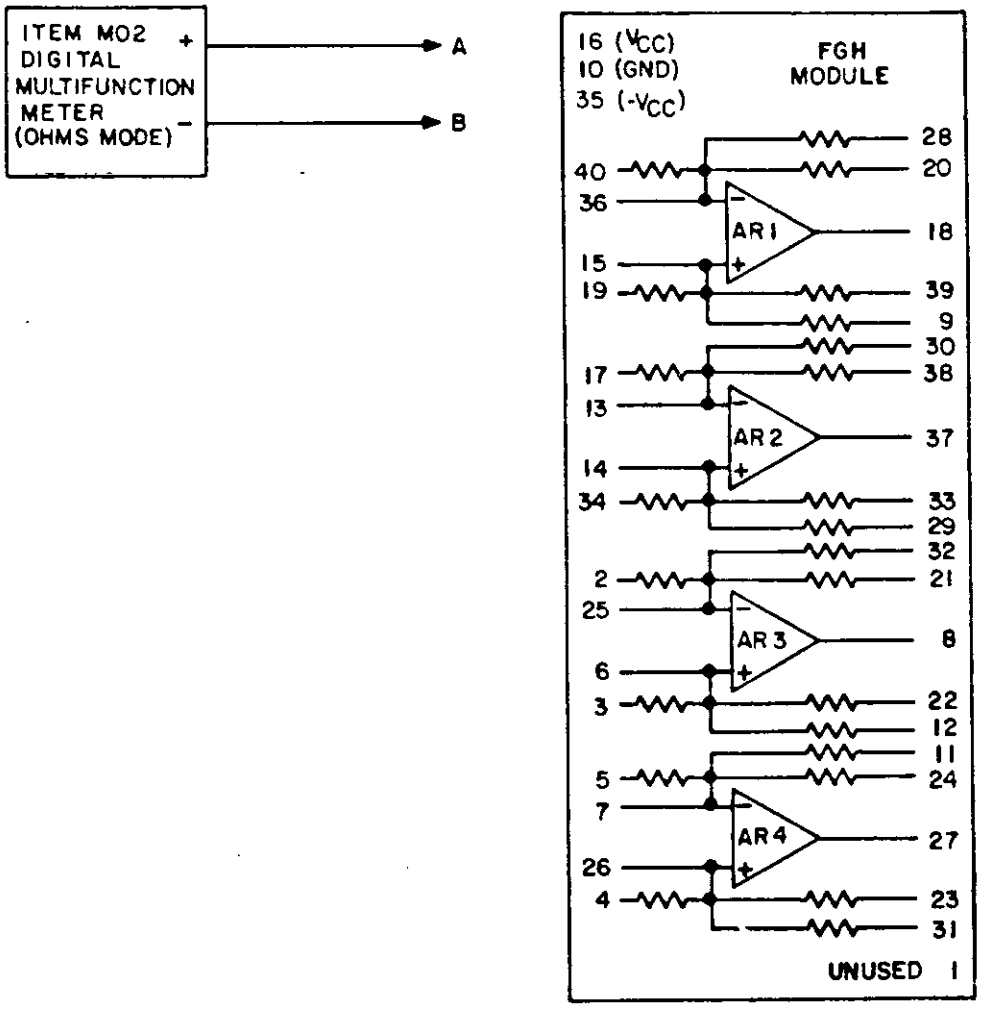


FIGURE 10. Resistance test circuit for table VII.

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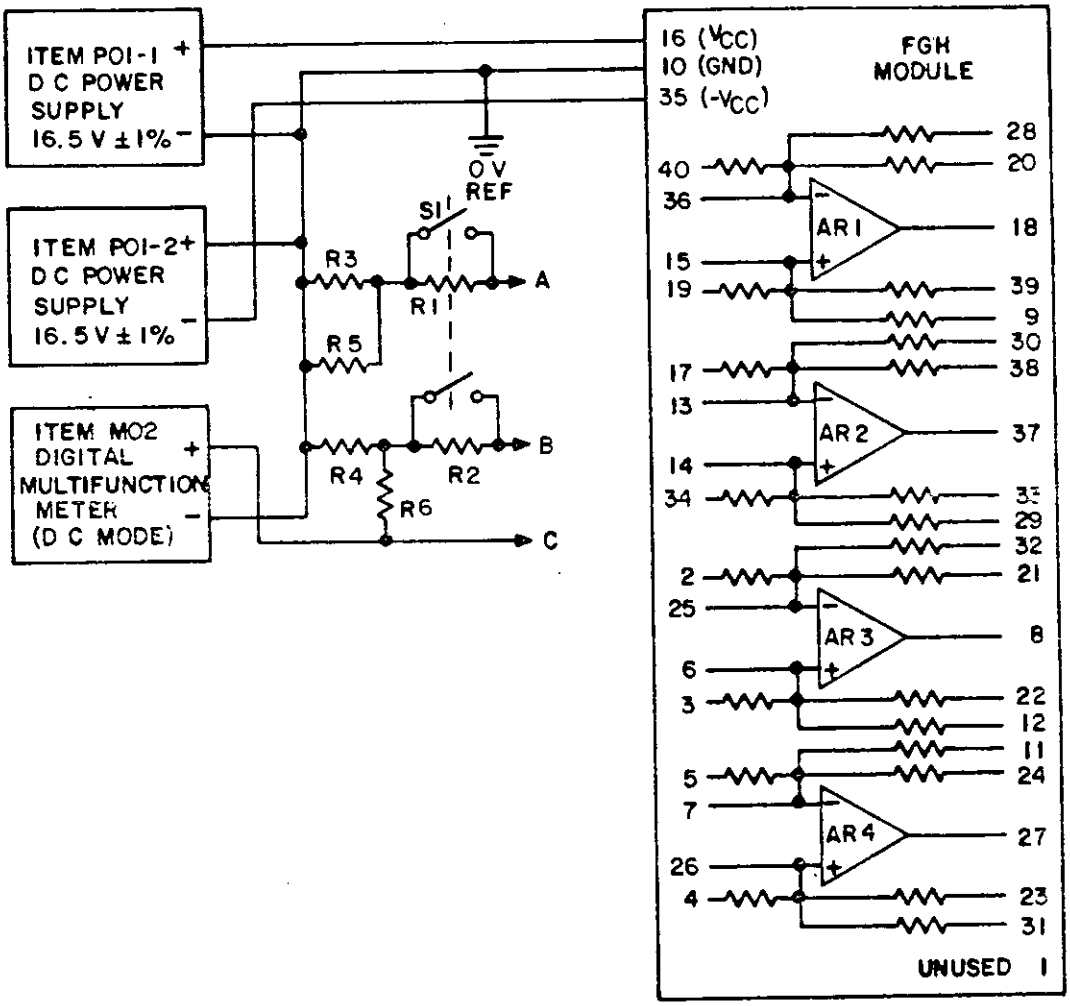
TABLE VIII. Input offset voltage and input offset current check test. 1/ 2/

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	Input offset voltage check	15	36	18	-300	300	mV	11
G2	↓	14	13	37	↓	↓	↓	↓
G3		6	25	8				
G4		26	7	27				
H1	Input offset current check	15	36	18	-30	30		
H2	↓	14	13	37	↓	↓		
H3		6	25	8				
H4		26	7	27				

1/ $$V_{IO} = \frac{V_{OUT}}{\frac{R6 + R4}{R4}} \approx \frac{V_{OUT}}{100}$$

2/ $$I_{IO} = \frac{\Delta V_{OUT}}{\text{Gain} \times R_{Series}} = \frac{\text{(Test condition G) - (Item M02 reading)}}{100 \times 10\ k\Omega}$$

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NOTES:
1. R1 = R2 = 10 kΩ ±0.1%, 1/8 W.
2. R3 = R4 = 49.9Ω ±1%, 1/8 W.
3. R5 = R6 = 5 kΩ ±1%, 1/8 W.
4. Switch S1 (DPST) position: Closed for test conditions G1-G4 and open for test conditions H1-H4.

FIGURE 11. Input offset voltage and input offset current check test circuit for table VIII.

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TABLE IX. Input bias current check test. 1/ 2/

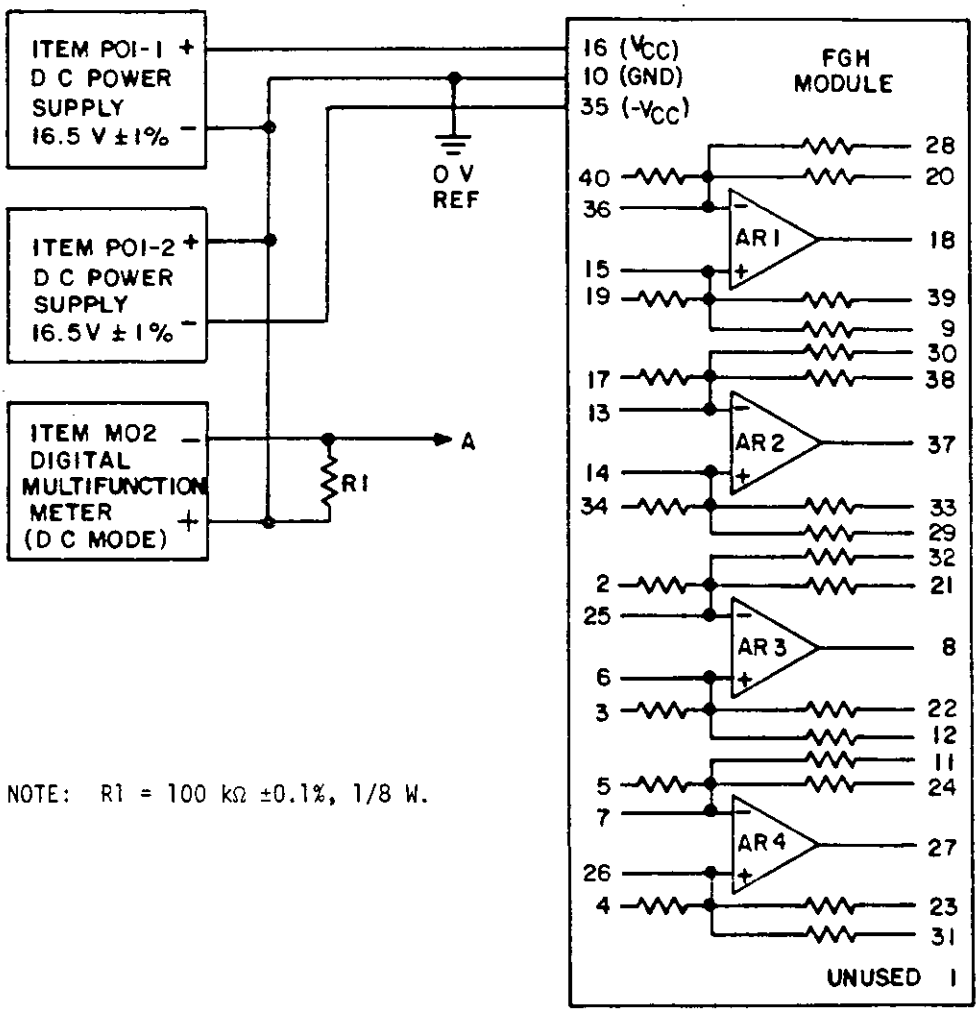
Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Input bias current check	36,15	0.2	22.0	mV	12
J2	↓	13,14	↓	↓	↓	↓
J3		25, 6				
J4		7,26				

1/ Average input bias current (table I) = $\frac{\text{Test condition J}}{100\ \text{k}\Omega \times 2}$

2/ This test in conjunction with test condition H (input offset current) verifies an input bias current ≤ 125 nA per input at 25°C initial where:

$$I_{IB}\ (\text{max}) = I_{IB}\ (\text{Avg}) + \frac{|I_{IO}|}{2}$$

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NOTE: R1 = 100 kΩ ±0.1%, 1/8 W.

FIGURE 12. Input bias current check test circuit for table IX.

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TABLE X. AC gain functional check, small signal bandwidth and gain peaking check test. 1/ 2/

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	AC gain functional check	40	28,18	39,9	4.85	5.10	V rms	13
K2		17	30,37	33,39				
K3		2	32,8	22,12				
K4		5	11,27	23,31				
K5		40	20,18	39,19	0.980	1.020		
K6		17	38,37	33,34				
K7		2	21,8	22,3				
K8		5	24,27	23,4				
L1	Small signal bandwidth	15	36,18	Open	0.438	5.00	MHz	
L2		14	13,37					
L3		6	25,8					
L4		26	7,27					
M1	Gain peaking check	15	36,18		19.98	27	mV rms	
M2		14	13,37					
M3		6	25,8					
M4		26	7,27					

1/ Gain = $\frac{-V_{OUT}}{V_{IN}} = \frac{\text{-Test condition K}}{\text{0.5 V rms}}$

2/ Gain peaking ratio (table I) = $\frac{V_{OUT}\ \text{(maximum)}}{V_{IN}} = \frac{\text{Test condition M}}{\text{20 mV rms}}$

This test guarantees a phase margin of $\approx$ 45°. The minimum limit allows for the -0.1% input tolerance.

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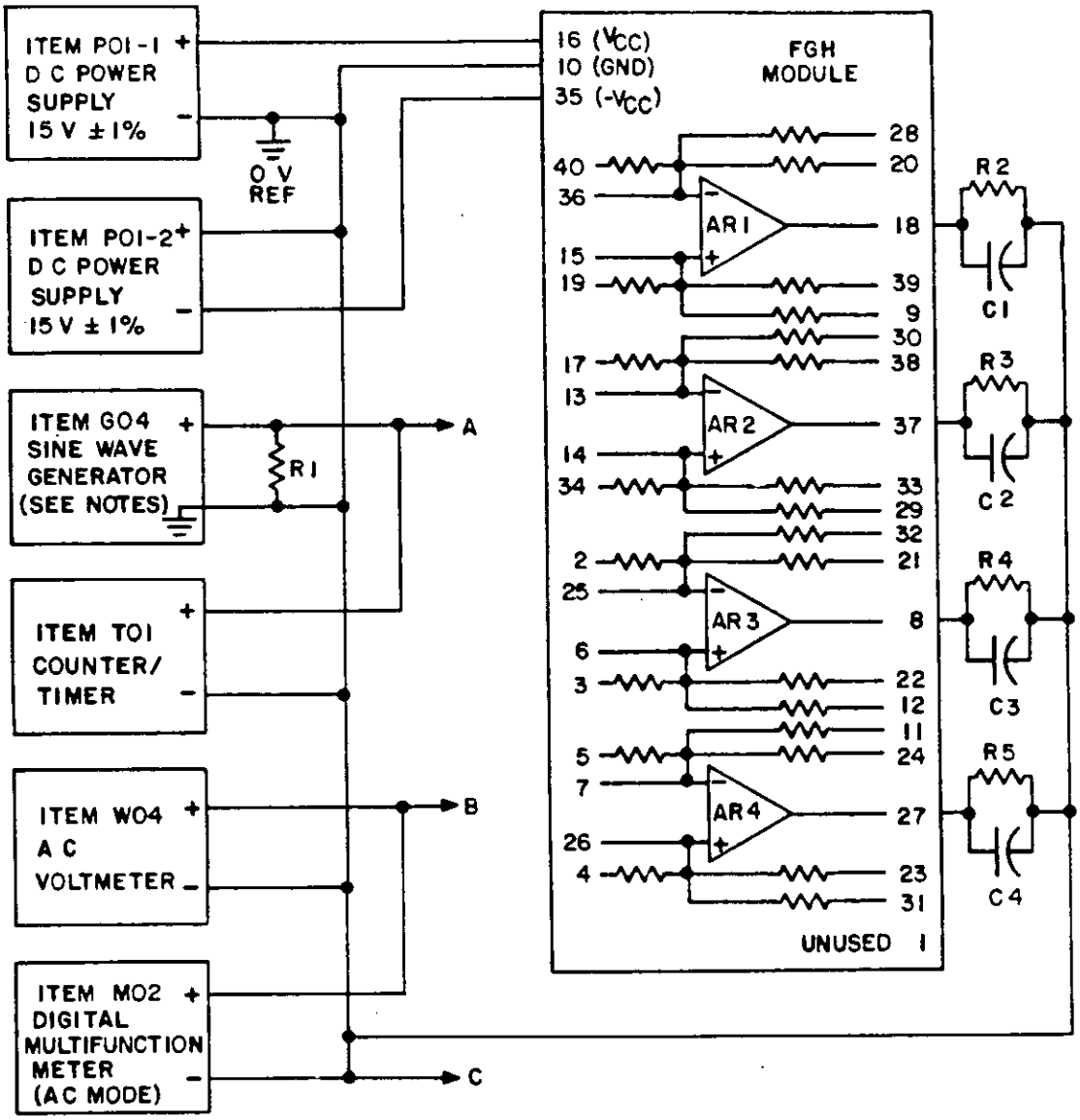


FIGURE 13. AC gain functional, small signal bandwidth and gain peaking check test circuit for table X.

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NOTES:
1. R1 = 49.9Ω ±5%, 1 W.
2. R2-R5 = 2 kΩ ±1%, 1/8 W.
3. C1-C4 = 100 pF ±20%.

4. a. Adjust item G04 to yield 0.5 Vrms ±0.1%, 1.0 kHz ±1% on item M02 at point A with A connected.
   b. With B connected per test condition K1, take readings on item M02.
   c. Repeat for K2-K4.
   d. Repeat notes 4a-4c for K5-K8 except adjust item G04 for 1 Vrms ±0.1%.

5. a. Adjust item G04 to yield 20 mVrms ±1%, 1.0 kHz ±1% at point B with B connected per test condition L1.
   b. For test condition L1 maintain input voltage within ±1% while increasing the frequency until output rolls off to 14.14 mVrms (-3 dB).
   c. Read bandwidth on item T01.
   d. Repeat for test conditions L2-L4.

6. a. For test condition M1, connect as per table X.
   b. Sweep frequency from 1.0 kHz to 5 MHz while maintaining input voltage at point A at 20 mVrms ±0.1%.
   c. Using item W04, note and record maximum voltage at point B (reading M1).
   d. Repeat for test conditions M2, M3, and M4.

FIGURE 13. AC gain functional, small signal bandwidth and gain peaking check test circuit for table X - Continued.

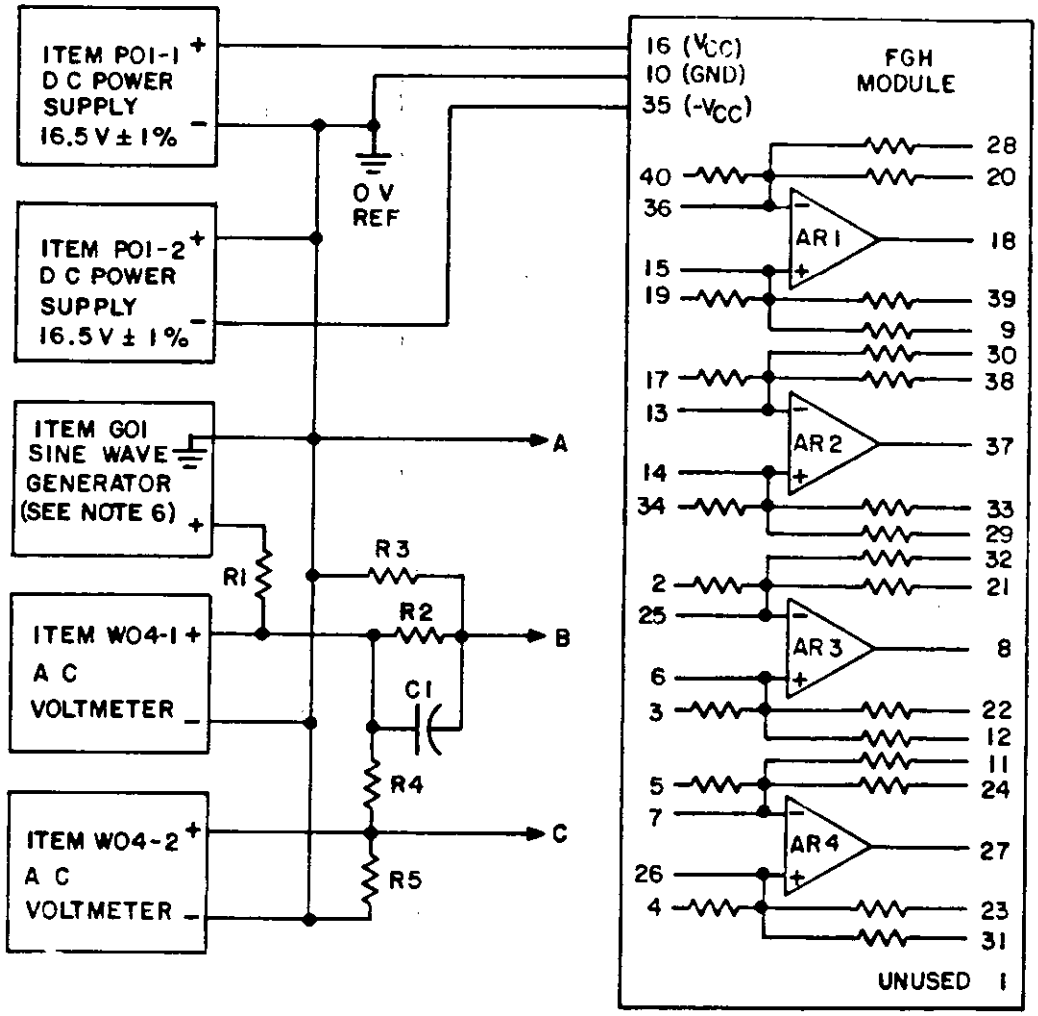
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TABLE XI. Large signal voltage gain check test. 1/

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	Large signal voltage gain check	15	36	18	1	14	mV rms	14
N2	↓	14	13	37	↓	↓	↓	↓
N3		6	25	8				
N4		26	7	27				

1/ Large signal voltage gain $\triangleq \frac{7 \text{ V rms}}{\text{Test condition N}} \times 100$

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NOTES:
1. R1 = R4 = 49.9 kΩ ±1%, 1/8 W.
2. R2 = 100 kΩ ±1%, 1/8 W.
3. R3 = 1.0 kΩ ±1%, 1/8 W.
4. R5 = 10 kΩ ±1%, 1/4 W.
5. C1 = 470 pF ±10%, 50 Vdc.
6. For each test condition:

   a. Adjust item G01 frequency to 10 Hz ±5% and amplitude to 7.0 Vrms ±1% at point C as measured on item W04-2 with points B and C connected.
   b. Read item W04-1.

7. Since items W04-1 and W04-2 are not used simultaneously, only one item W04 is required.
8. Test condition N verifies a minimum large signal voltage gain at 25°C initial of 50,000.

FIGURE 14. Large signal voltage gain check test circuit for table XI.

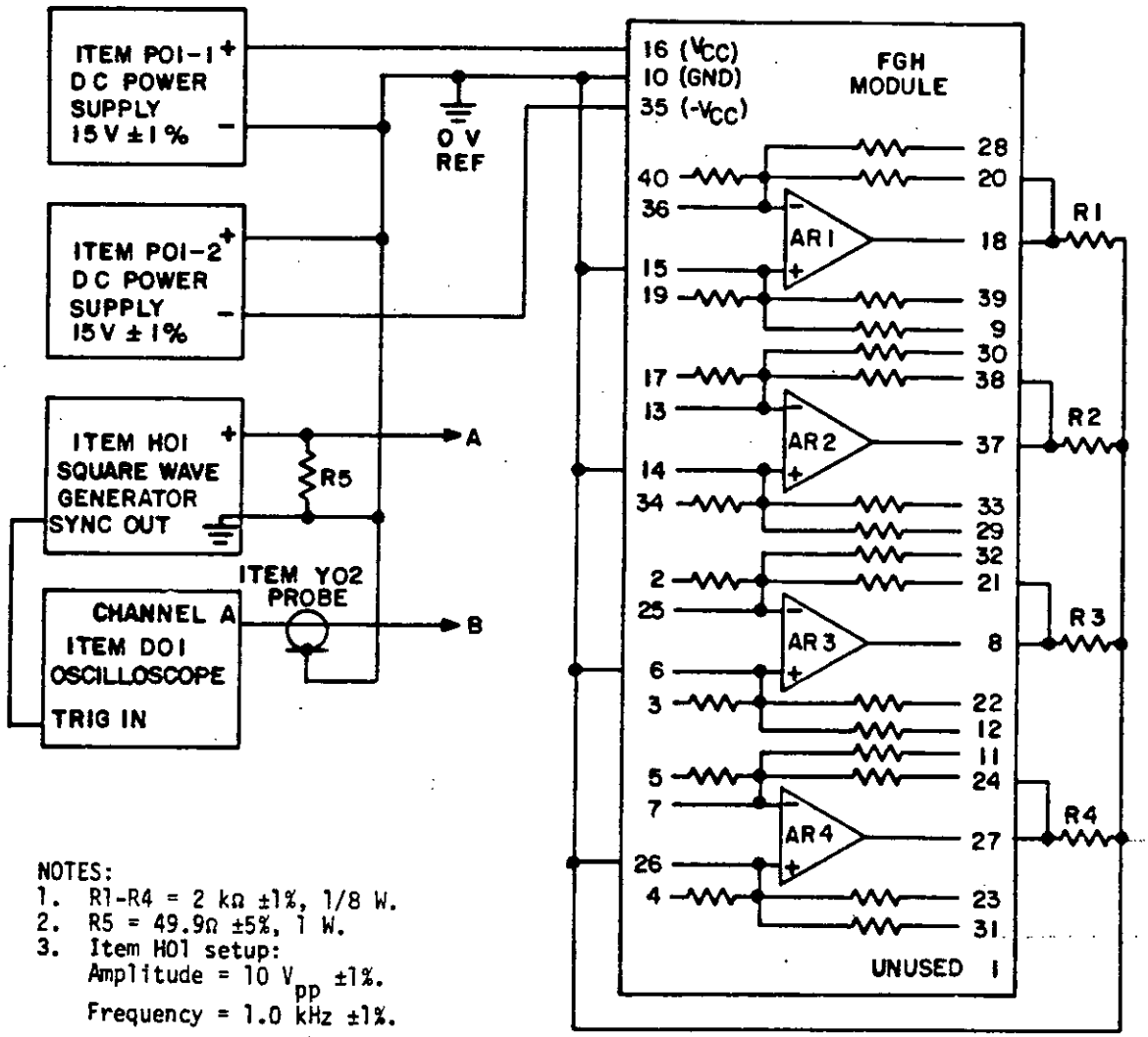
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TABLE XII. Output slew rate check test. 1/

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
P1	Output slew rate check (negative)	40	18	0.4	13.3	µs	15,16
P2	Output slew rate check (positive)	40	18				
P3	Output slew rate check (negative)	17	37				
P4	Output slew rate check (positive)	17	37				
P5	Output slew rate check (negative)	2	8				
P6	Output slew rate check (positive)	2	8				
P7	Output slew rate check (negative)	5	27				
P8	Output slew rate check (positive)	5	27				

1/ Slew rate = $\frac{4\ V}{\text{Test condition P}}$

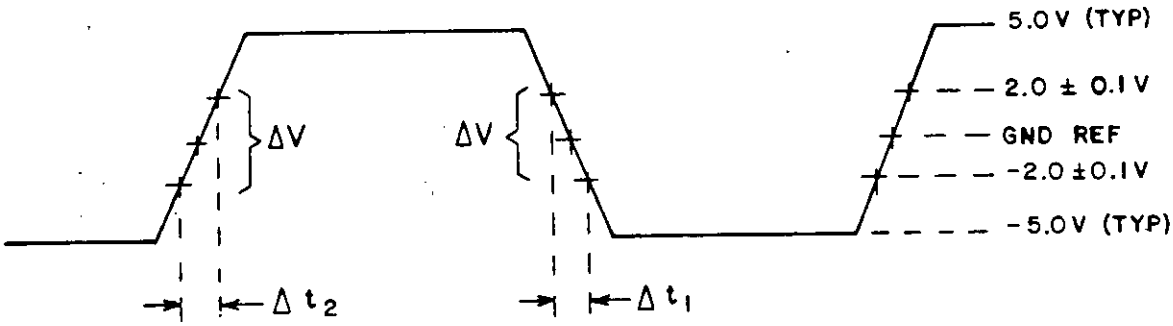
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NOTES:
1. R1-R4 = 2 kΩ ±1%, 1/8 W.
2. R5 = 49.9Ω ±5%, 1 W.
3. Item HO1 setup:
   Amplitude = 10 V_{pp} ±1%.
   Frequency = 1.0 kHz ±1%.

FIGURE 15. Output slew rate check test circuit for table XII.

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WAVEFORM AT POINT B

NOTES:
1. $\Delta V/_{\Delta t}$ = Output slew rate.
2. Δt_1 = P1-P7 (odd), negative slew rate.
   Δt_2 = P2-P8 (even), positive slew rate.

FIGURE 16. Output slew rate check test circuit waveform for table XII.

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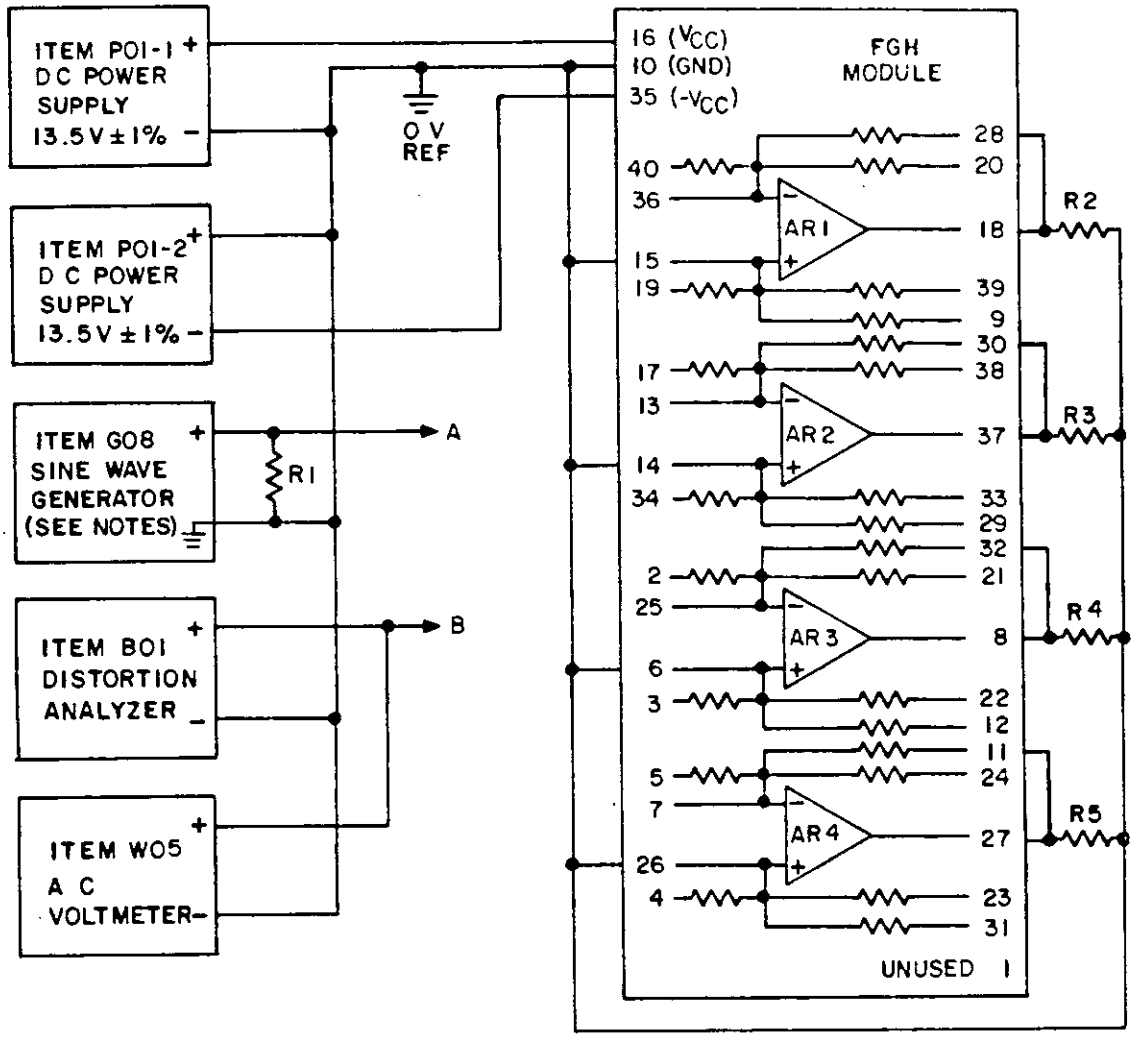
TABLE XIII. Maximum output voltage test. 1/ 2/

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	Maximum output voltage	40	18	6.00	9.55	V rms	17
R2	↓	17	37	↓	↓	↓	↓
R3		2	8				
R4		5	27				

1/ This test assumes that the item G08 output has no significant distortion.

2 Test condition R verifies a minimum voltage swing of ±8.5 V_{peak} and a minimum output drive capability of 4.25 mA at distortion ≤ 1 percent.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2-R5 = 2 kΩ ±1%, 1/8 W.
3. Item G08 setup:
   Frequency = 1 kHz ±1%.
   Amplitude = 0.5 Vrms ±10%.
4. a. Make connections per table XIII, test condition R1.
   b. Increase amplitude of item G08 until distortion (on item B01) equals 1 ±0.1%.
   c. Record item W05 reading as test condition R1 value and repeat 4a - 4c for test condition R2-R4.

FIGURE 17. Maximum output voltage test circuit for table XIII.

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TABLE XIV. Common mode rejection ratio check test. 1/

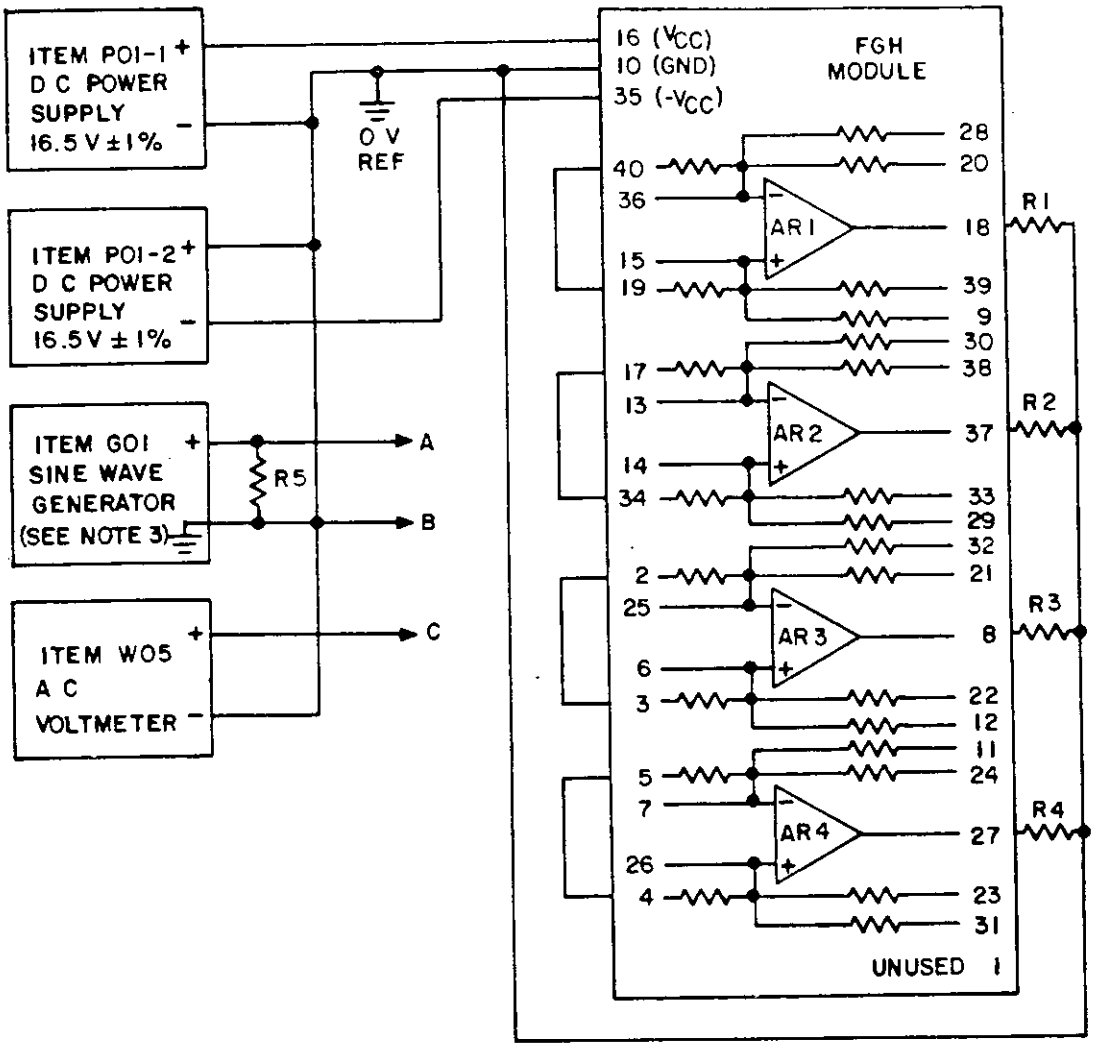
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	Common mode rejection ratio check	40	39	18,20	0	10	mV rms	18
S2		17	33	37,38				
S3		2	22	8,21				
S4		5	23	24,27				
S5		40	9	18,28		100		
S6		17	29	30,37				
S7		2	12	8,32				
S8		5	31	11,27				

1/ $CMRR = 20 \log \left(\frac{V_{OUT}/gain}{V_{IN}} \right) = 20 \log \left(\frac{\text{Item W05 reading/gain}}{1 \text{ V rms}} \right)$

Gain = 1 for test conditions S1-S4

= 10 for test conditions S5-S8.

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NOTES:
1. R1-R4 = 2 kΩ ±1%, 1/8 W.
2. R5 = 49.9Ω ±5%, 1 W.
3. Item G01 setup:
   Amplitude = 1.0 Vrms ±1% (with point A connected).
   Frequency = 1.0 kHz ±1%.

FIGURE 18. Common mode rejection ratio check test circuit for table XIV.

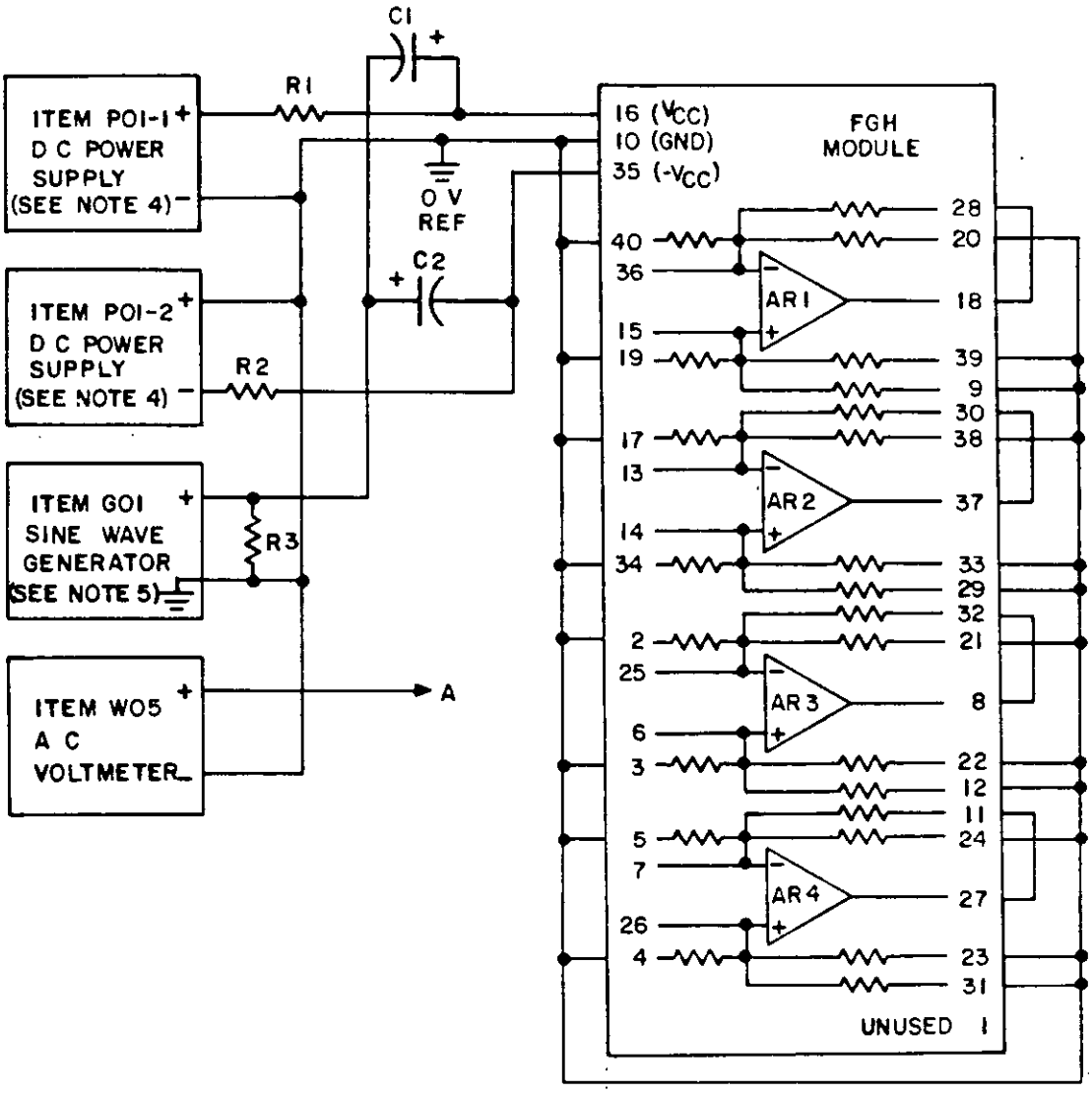
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TABLE XV. Power supply rejection ratio check test. 1/

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
T1	Power supply rejection ratio check	18	0	50	mV rms	19
T2	↓	37	↓	↓	↓	↓
T3		8				
T4		27				

1/ $PSRR = 20 \log\left(\frac{V_{OUT}/gain}{V_{IN}}\right) = 20 \log\left(\frac{\text{Item W05 reading}/20}{1 \text{ V rms}}\right)$

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NOTES:
1. R1 = R2 = 100Ω ±5%, 1/4 W.
2. R3 = 49.9Ω ±5%, 1 W.
3. C1 = C2 = 20 µF ±20%, 20 Vdc, electrolytic.
4. Adjust item P01-1 to obtain 15 V ±1% at pin 16; adjust item P01-2 to obtain 15 V ±1% at pin 35.
5. Adjust item G01 for 1 Vrms ±1%, 1.0 kHz ±1% at pins 16 and 35.

FIGURE 19. Power supply rejection ratio check test circuit for table XV.

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TABLE XVI. Circuit-to-circuit phase shift test.

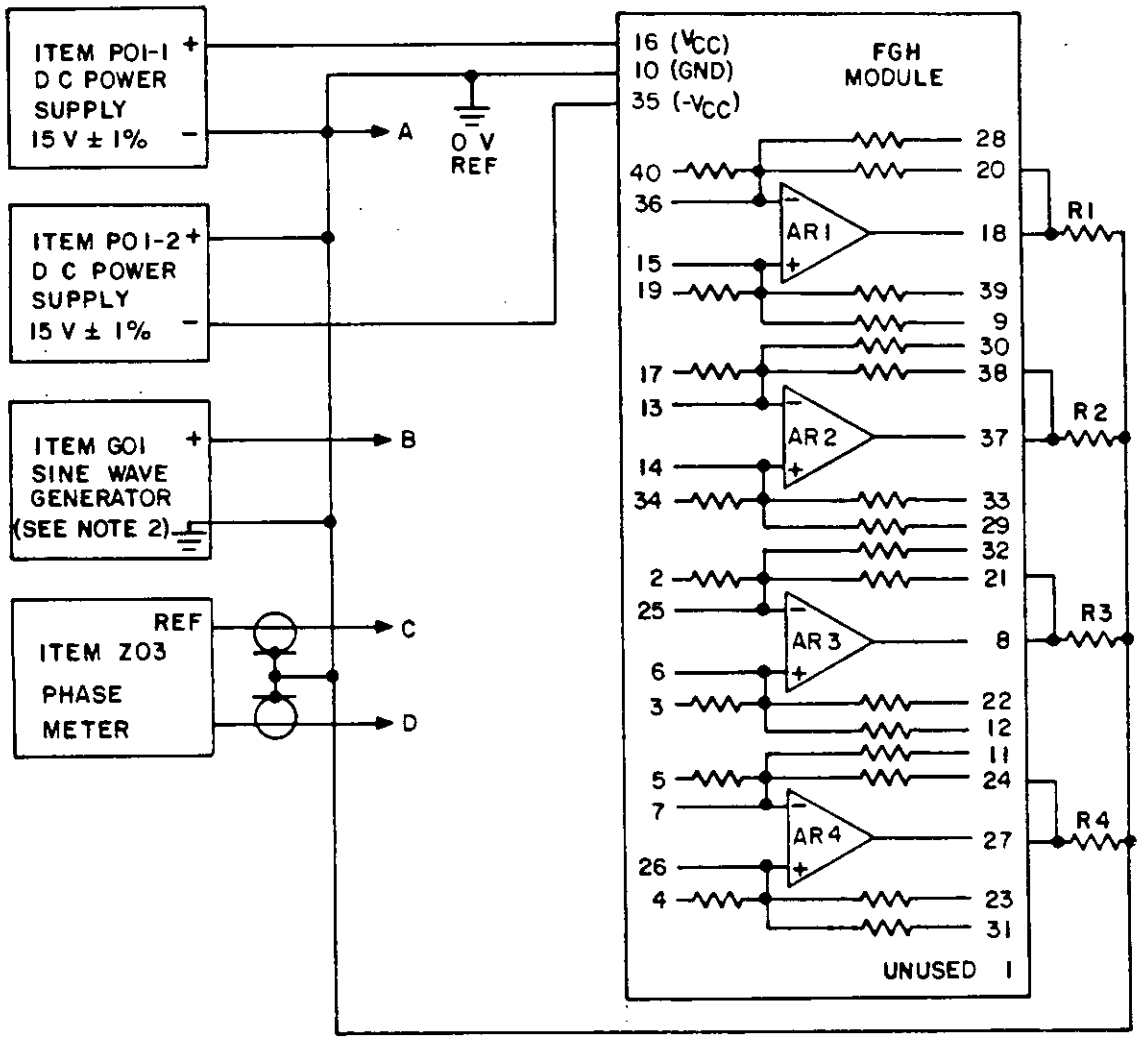
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	Circuit-to-circuit phase shift	14,15	17,40	18	37	-0.5	0.5	Degree	20
U2	Circuit-to-circuit phase shift	6,26	2,5	8	27	-0.5	0.5	Degree	20

TABLE XVII. Crosstalk check test. 1/

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	Crosstalk check	40	17,2,5	18	0	70	mV rms	21
V2	↓	17	40,2,5	37	↓	↓	↓	↓
V3		2	40,17,5	8				
V4		5	40,17,5	27				

1/ Crosstalk = $20 \log\left(\frac{V_{OUT}}{V_{IN}}\right) = 20 \log\left(\frac{\text{Item W05 reading}}{7.0 \text{ V rms}}\right)$.

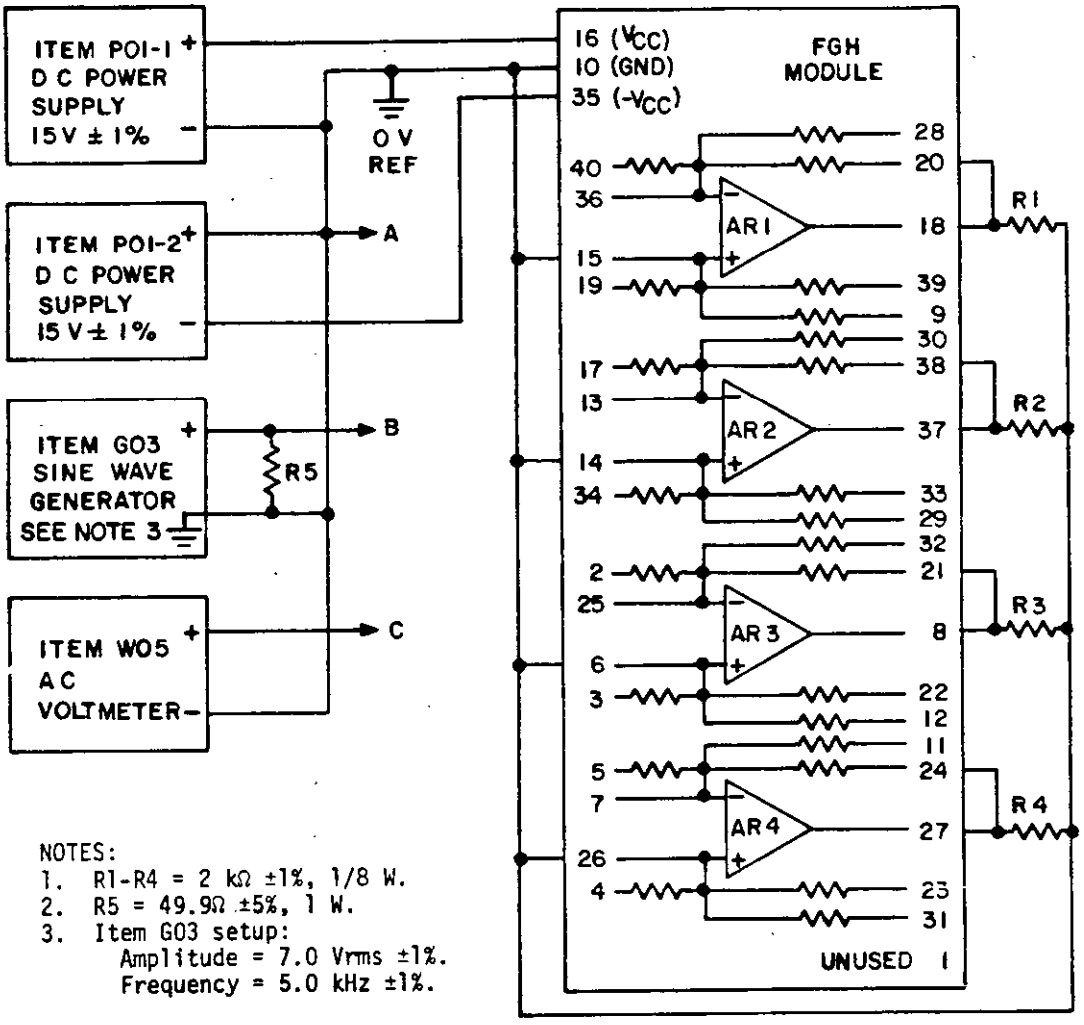
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NOTES:
1. R1-R4 = 2 kΩ ±1%, 1/8 W.
2. Item G01 setup:
   Amplitude = 0.5 Vrms ±1%.
   Frequency = 10 kHz ±1%.

FIGURE 20. Circuit-to-circuit phase shift test circuit for table XVI.

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NOTES:
1. R1-R4 = 2 kΩ ±1%, 1/8 W.
2. R5 = 49.9Ω ±5%, 1 W.
3. Item GO3 setup:
   Amplitude = 7.0 Vrms ±1%.
   Frequency = 5.0 kHz ±1%.

FIGURE 21. Crosstalk check test circuit for table XVII.

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